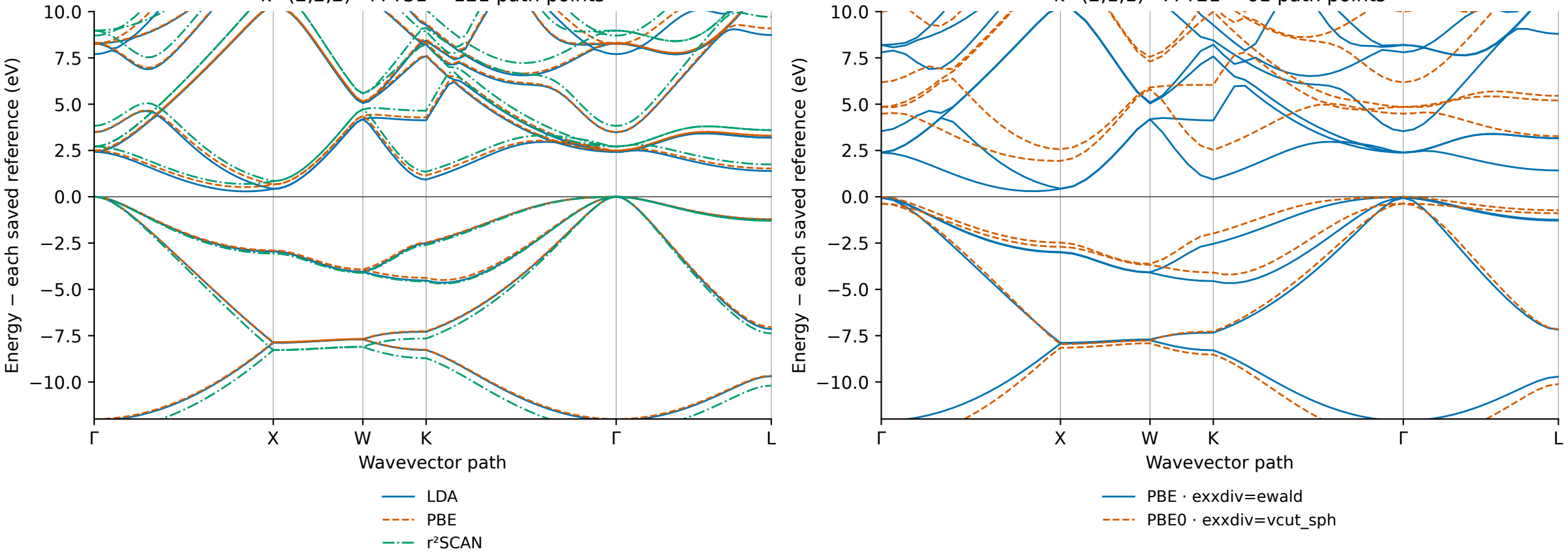
Silicon · all five saved KS/GKS cases and both direct GW data sets
GTH-DZVP-MOLOPT-SR / GTH-PBE · no spin–orbit coupling (upper panels); GW uses GTH-SZV (lower panels)

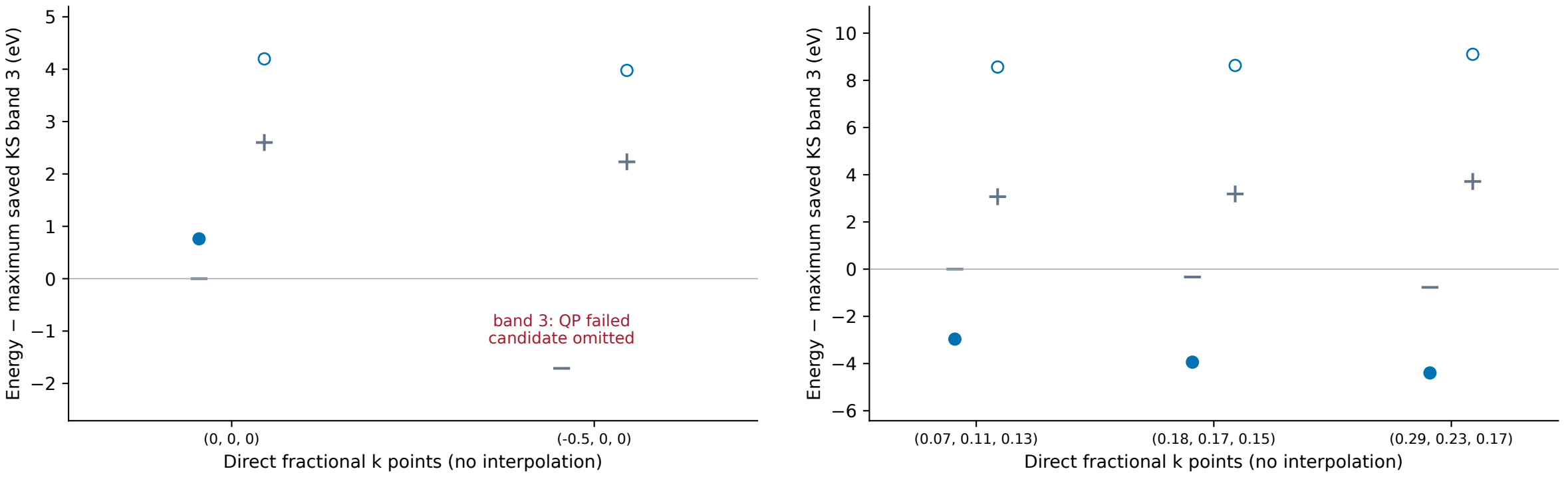
(a) Matched semilocal models
k=(2,2,2) · FFT31³ · 121 path points

(b) Compact PBE / PBE0 models
k=(2,1,1) · FFT21³ · 61 path points



(c) Original mesh: 3/4 QPs accepted
legacy folded OV · fc=True · nw=8

(d) Generic queries: 6/6 QPs accepted
full OV+VO · fc=False · nw=16



Upper panels: all five independent same-input band checks passed. Lower panels: QP solver certificates only; no independent Si GW benchmark.
GW: PBE W_0, source k=(2,1,1), FFT21³, Ewald exchange, η=0.001 Ha. The two GW prescriptions are distinct.
Saved finite models; basis, k mesh, FFT grid and vacuum convergence are not established.